

Class 18: Pertussis and the CMI-PB Project

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Section 1. Investigating pertussis cases by year

Q1. With the help of the R “addin” package datapasta assign the CDC pertussis case number data to a data frame called `cdc` and use **ggplot** to make a plot of cases numbers over time.

Note: Using `read.csv` instead, upon instruction that it is okay to do so in place of `datapasta`.

```
# Read in cdc pertussis data
cdc <- read.csv("U.S. Reported Pertussis Cases_ 1922 - 2025.csv")
```

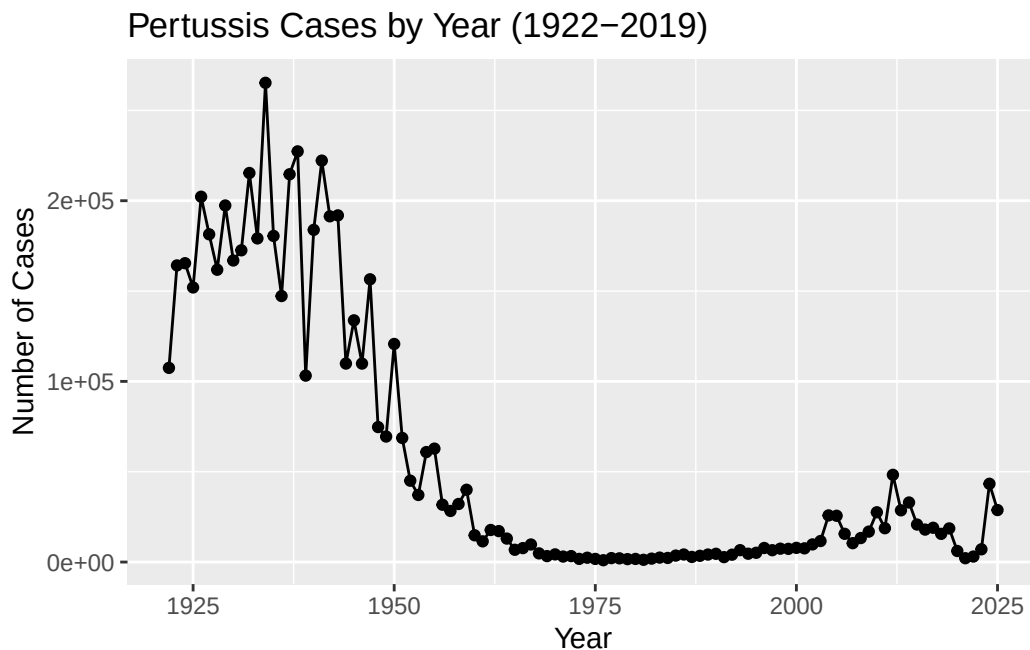
```
# Convert pertussis case numbers from strings to numerics

cdc$Number.of.Reported.Pertussis.Cases <- as.numeric(gsub(",", "" , 
                                                    cdc$Number.of.Reported.Pertussis.C
```

```
# Plot case numbers over time

library(ggplot2)
```

```
ggplot(cdc) +
  aes(x = Year, y = Number.of.Reported.Pertussis.Cases) +
  geom_point() +
  geom_line() +
  labs(title = "Pertussis Cases by Year (1922-2019)", x = "Year", y = "Number of Cases")
```



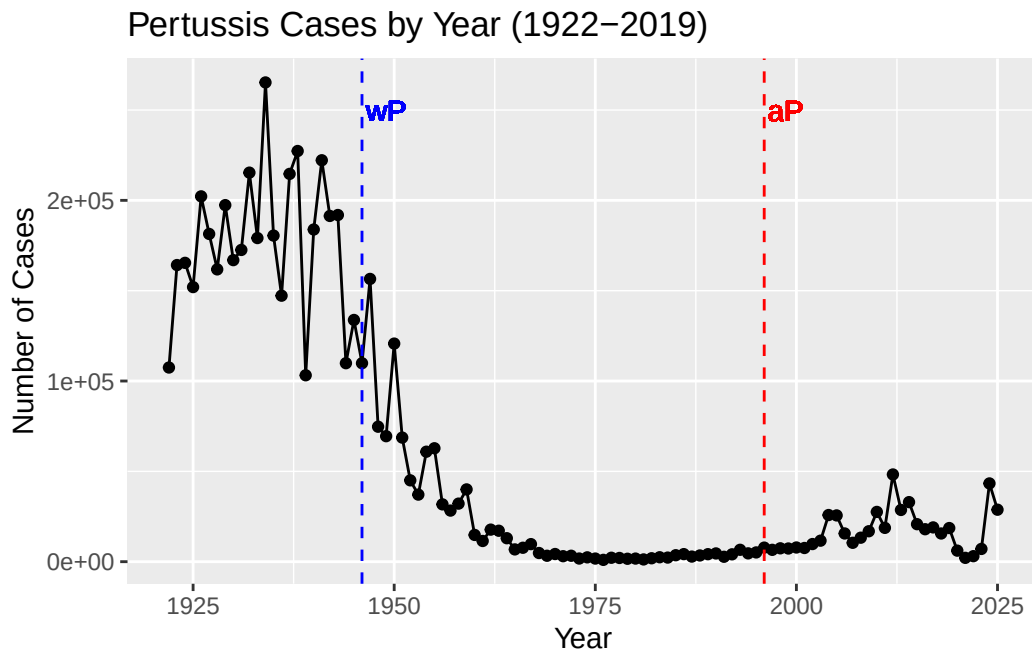
Section 2. A tale of two vaccines (wP & aP)

Q2. Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

```
# Add lines for wP introduction and switch to aP

ggplot(cdc) +
  aes(x = Year, y = Number.of.Reported.Pertussis.Cases) +
  geom_point() +
  geom_line() +
  labs(title = "Pertussis Cases by Year (1922-2019)", x = "Year", y = "Number of Cases") +
  geom_vline(xintercept = 1946, color = "blue", linetype = "dashed") +
  geom_vline(xintercept = 1996, color = "red", linetype = "dashed") +
```

```
geom_text(x = 1946, y = 250000, label = "wP", color = "blue", hjust = -0.1) +  
geom_text(x = 1996, y = 250000, label = "aP", color = "red", hjust = -0.1)
```



It appears that there was a substantial decline in pertussis cases following the introduction of the wP vaccine, in 1946. Though levels of pertussis cases initially remained steadily low past 1975, a shift occurred following the introduction of the aP vaccine: cases are shown to rise again following 1996.

Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

Pertussis cases seem to have been *increasing* since the introduction of the aP vaccine in 1996. This is in contrast to the downward trend of cases after the original, wP vaccine's introduction.

It is possible that the increase in cases is due to the aP vaccine granting a shorter period of protection/immunity from pertussis, as compared to the wP vaccine. Thus, the immunity of younger generations vaccinated with aP may be reducing/waning sooner following childhood vaccination, resulting in increased susceptibility to pertussis and more pertussis cases. That being said, the following are also possible: diagnosis methods have improved (i.e. better or more accessible testing), fewer people are getting vaccinated (i.e. due to vaccine hesitancy, reduced caution, etc.), or the pertussis bacteria has evolved.

Section 3. Exploring CMI-PB data

```
# Load jsonlite package to read, write and process JSON data
library(jsonlite)
```

```
# Read in metadata about individuals in study
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)

# Take a quick look at the metadata
head(subject, 3)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	2	wP	Female	Not Hispanic or Latino	White
3	3	wP	Female	Unknown	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

There are 87 aP vaccinated and 85 wP vaccinated subjects.

Q5. How many Male and Female subjects/patients are in the dataset?

```
table(subject$biological_sex)
```

```
Female  Male
112     60
```

There are 60 Male subjects and 112 Female subjects.

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

```
table(subject$biological_sex, subject$race)
```

	American Indian/Alaska Native	Asian	Black or African American
Female	0	32	2
Male	1	12	3

	More Than One Race	Native Hawaiian or Other Pacific Islander
Female	15	1
Male	4	1

	Unknown or Not Reported	White
Female	14	48
Male	7	32

The breakdown by race and biological sex is shown in the table above. It appears the most represented groups are White females (48), followed by White males and Asian females (these last two groups are tied in count, 32).

There is comparatively much less representation of the American Indian/Alaska Native, Black/African American, and Pacific Islander races. There is also higher representation of females than males.

Side-Note: Working with dates

```
# Example
```

```
library(lubridate)
```

```
# Today's date
```

```
today()
```

```
[1] "2026-06-01"
```

```
# Days since new year 2000
```

```
today() - ymd("2000-01-01")
```

Time difference of 9648 days

```
# Convert above answer to years
time_length(today() - ymd("2000-01-01"), "years")
```

```
[1] 26.41478
```

Today is “2026-05-28”. 9644 days have passed since new year 2000, which is 26.4 years.

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```
# Store age (in days) of subjects
subject$age <- today() - ymd(subject$year_of_birth)
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
# Filter to wP-vaccinated subjects.
wp <- subject |>
  filter(infancy_vac == "wP")

# Examine 6 number summary of their ages.
round ( summary (time_length(wp$age, "years")) )
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	33	35	37	40	58

(i) The average age of wP individuals is 37 years.

```
# Filter to aP-vaccinated subjects.
ap <- subject |>
  filter(infancy_vac == "aP")

# Examine 6 number summary of their ages.
round ( summary (time_length(ap$age, "years")) )
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	27	28	28	29	35

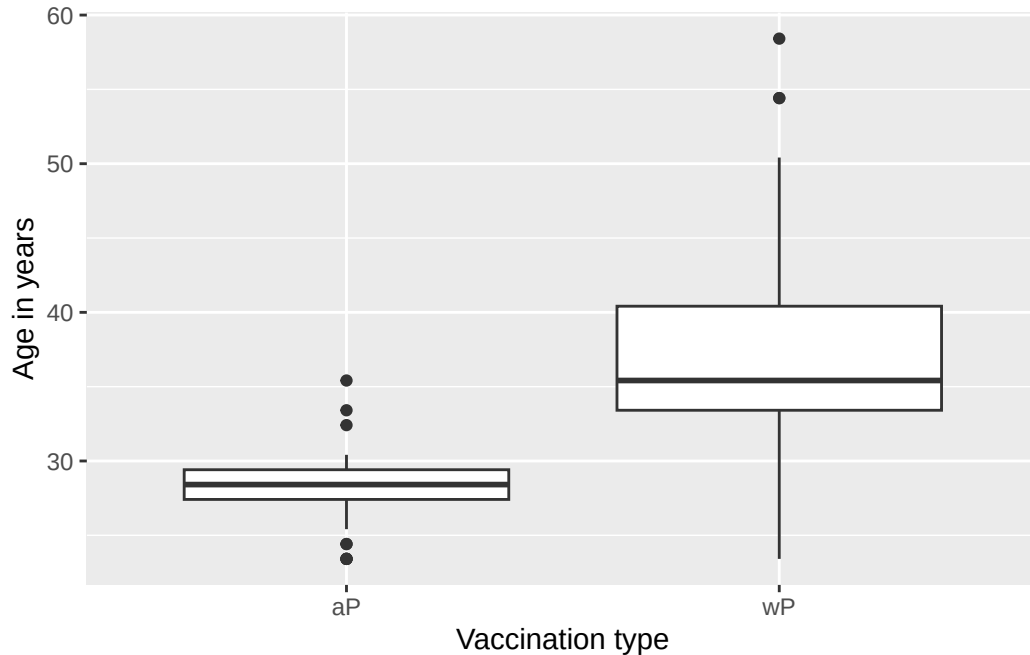
(ii) The average age of aP individuals is 28 years.

```
# Run t.test of wP and aP ages to determine if there's a stat difference
t.test(time_length(wp$age, "years"), time_length(ap$age, "years"))
```

Welch Two Sample t-test

```
data: time_length(wp$age, "years") and time_length(ap$age, "years")
t = 12.918, df = 104.03, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 7.407351 10.094058
sample estimates:
mean of x mean of y
37.06084 28.31013
```

```
ggplot(subject) +
  aes(x = as.factor(infancy_vac), y = time_length(age, "years")) +
  geom_boxplot() +
  xlab("Vaccination type") +
  ylab("Age in years")
```



(iii) Results from a two-sample t test of the ages of wP-vaccinated subjects and aP-vaccinated subjects indicate that the difference in ages *is statistically significant*. The p-value is very low ($2.2e-16$), so we find evidence for the alternative hypothesis that the true difference in mean ages is not equal to 0. On average, the wP individuals (mean age = 37 years) are older than the aP individuals (mean age = 28 years).

Even from the summary stats and a representative boxplot, we observe that the middle 50% of the wP group (i.e. 1st quartile age to 3rd quartile age) is 33-40 years old, whereas the middle 50% of the aP group is younger at 27-29 years old.

Q8. Determine the age of all individuals at time of boost?

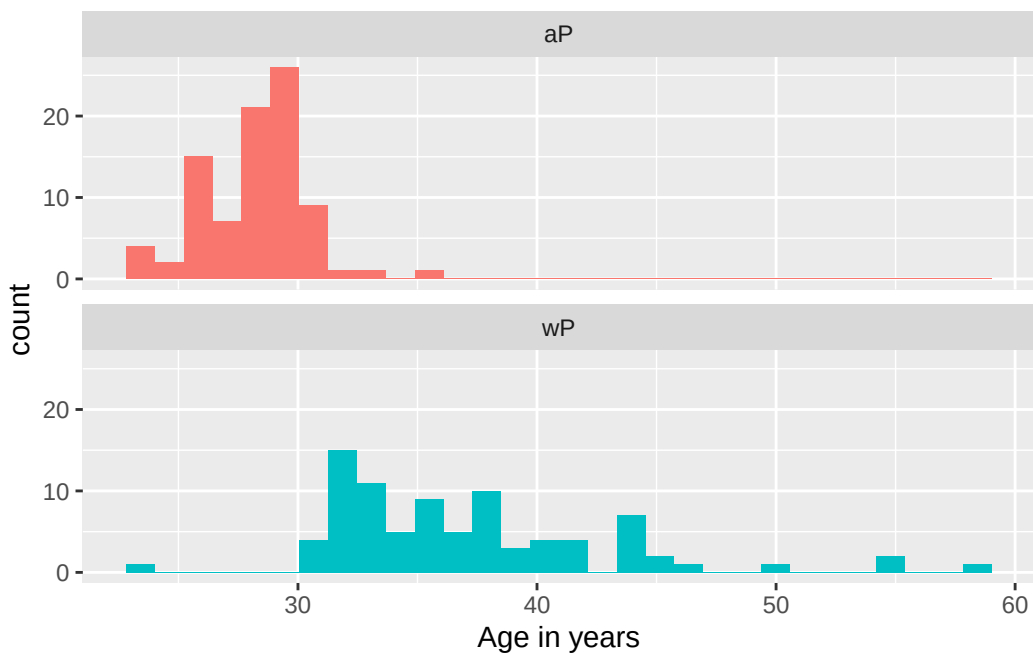
```
age_at_boost <- time_length( ymd(subject$date_of_boost) - ymd(subject$year_of_birth), "years")
head(age_at_boost)
```

```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?


```
ggplot(subject) +
  aes(time_length(age, "year"),
      fill=as.factor(infancy_vac)) +
  geom_histogram(show.legend=FALSE) +
  facet_wrap(vars(infancy_vac), nrow=2) +
  xlab("Age in years")
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



The two groups appear to have a significant difference in age: the wP-vaccinated group is older than the aP-vaccinated group. The distribution of ages for the wP group is shifted right on the histogram (toward higher ages), relative to the distribution for the aP group. Furthermore, the t-test conducted earlier found evidence for a difference, with a p-value of 2.2e-16.

Joining multiple tables

```
# Read the specimen and ab_titer tables into R
specimen <- read_json("https://www.cmi-pb.org/api/specimen", simplifyVector = TRUE)
titer <- read_json("https://www.cmi-pb.org/api/plasma_ab_titer", simplifyVector = TRUE)
```

Q9. Complete the code to join `specimen` and `subject` tables to make a new merged data frame containing all specimen records along with their associated subject details:

```
# Merge specimen and subject tables.
meta <- inner_join(specimen, subject)
```

Joining with ``by = join_by(subject_id)``

```
dim(meta)
```

```
[1] 1503    14
```

```
head(meta)
```

```
specimen_id subject_id actual_day_relative_to_boost
1           1           1                        -3
2           2           1                         1
3           3           1                         3
4           4           1                         7
5           5           1                        11
6           6           1                        32
planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
1                             0         Blood      1         wP         Female
2                             1         Blood      2         wP         Female
3                             3         Blood      3         wP         Female
4                             7         Blood      4         wP         Female
5                            14         Blood      5         wP         Female
6                            30         Blood      6         wP         Female
ethnicity race year_of_birth date_of_boost dataset
1 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
2 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
3 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
4 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
5 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
6 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
age
1 14761 days
2 14761 days
3 14761 days
```

```
4 14761 days
5 14761 days
6 14761 days
```

- Note: `inner_join` was used, to ensure all rows have both a `specimen_id` and an associated `subject_id`.

Q10. Now using the same procedure join `meta` with `titer` data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
abdata <- inner_join(titer, meta)
```

Joining with ``by = join_by(specimen_id)``

```
dim(abdata)
```

```
[1] 52576    21
```

Q11. How many specimens (i.e. entries in `abdata`) do we have for each isotype?

```
table(abdata$isotype)
```

```
 IgE   IgG  IgG1  IgG2  IgG3  IgG4
6698  5389 10117 10124 10124 10124
```

There are 6698 specimens of IgE, 5389 specimens of IgG, 10117 specimens of IgG1, and 10124 specimens each of IgG2, IgG3, and IgG4.

Q12. What are the different `$dataset` values in `abdata` and what do you notice about the number of rows for the most “recent” dataset?

```
table(abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
      31520         8085         7301         5670
```

`$dataset` has 4 values, representing the 4 independent datasets/cohorts created over 4 years: `2020_dataset`, `2021_dataset`, `2022_dataset`, `2023_dataset`. By creating a new, independent dataset each year, the researchers could use the prior years’ datasets as “training data” for their computational prediction models, and subsequently test the models on the newest dataset.

The most recent dataset, from 2023, has the fewest number of rows (5670, compared to 31520 in the first, 2020 dataset).

Section 4. Examine IgG Ab titer levels

```
# Filter abdata for IgG isotype
igg <- abdata %>% filter(isotype == "IgG")
head(igg)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgG	TRUE	PT	68.56614	3.736992
2	1	IgG	TRUE	PRN	332.12718	2.602350
3	1	IgG	TRUE	FHA	1887.12263	34.050956
4	19	IgG	TRUE	PT	20.11607	1.096366
5	19	IgG	TRUE	PRN	976.67419	7.652635
6	19	IgG	TRUE	FHA	60.76626	1.096457

	unit	lower_limit_of_detection	subject_id	actual_day_relative_to_boost
1	IU/ML	0.530000	1	-3
2	IU/ML	6.205949	1	-3
3	IU/ML	4.679535	1	-3
4	IU/ML	0.530000	3	-3
5	IU/ML	6.205949	3	-3
6	IU/ML	4.679535	3	-3

	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex
1	0	Blood	1	wP	Female
2	0	Blood	1	wP	Female
3	0	Blood	1	wP	Female
4	0	Blood	1	wP	Female
5	0	Blood	1	wP	Female
6	0	Blood	1	wP	Female

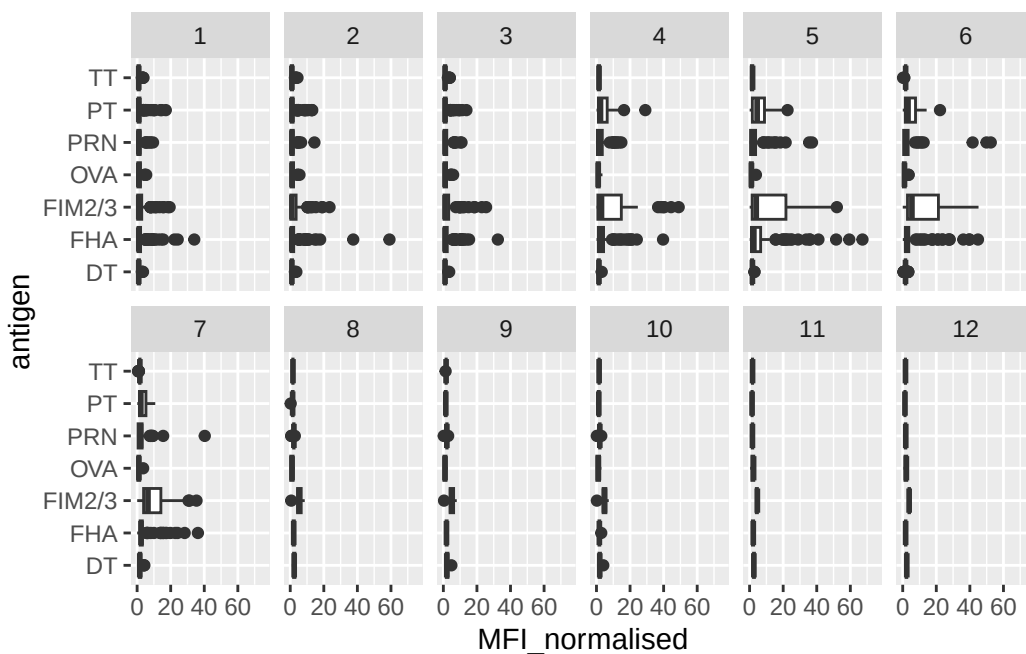
	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Unknown	White	1983-01-01	2016-10-10	2020_dataset
5	Unknown	White	1983-01-01	2016-10-10	2020_dataset
6	Unknown	White	1983-01-01	2016-10-10	2020_dataset

	age
1	14761 days
2	14761 days
3	14761 days
4	15857 days
5	15857 days
6	15857 days

Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens.

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
  xlim(0,75) +
  facet_wrap(vars(visit), nrow=2)
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

The antigens associated with differences over time include PT (pertussis toxin), PRN (pertactin autotransporter), FIM2/3 (Fimbrial protein 2/3), and FHA (Filamentous hemagglutinin). This makes sense since the booster vaccines use purified acellular pertussis antigens from the *Bordetella pertussis* bacterium—including PT, PRN, FIM2/3, and FHA—to trigger an immune response and pertussis-targeted antibody production.

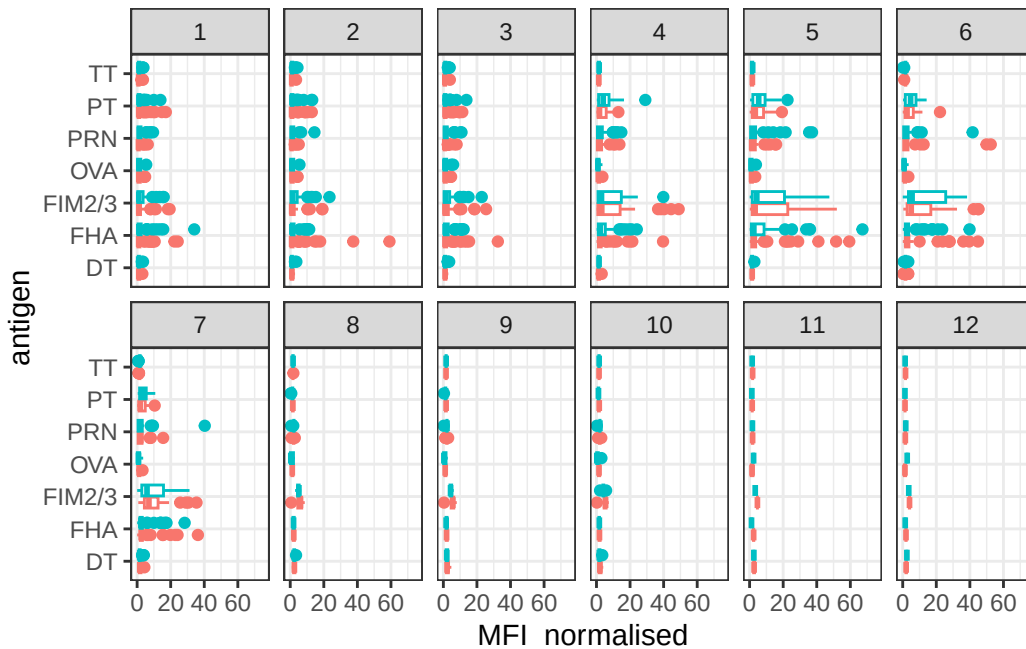
The OVA (ovalbumin) associated IgG antibody level does not increase since OVA is not present in the vaccine formulation, instead being used as a negative control in the assay. DT (diphtheria

toxoid) and TT (tetanus toxin), though also present in the booster vaccines along with the aP antigens, do not seem to show significant change in their associated IgG antibody titer levels. This may be due, in part to the reduced dosage of DT in the Tdap booster; additionally, the initial childhood vaccination may offer better long-term immunity/recognition against DT/TT than against pertussis, resulting in the Tdap booster triggering a stronger immune response to the pertussis-specific antigens.

```
# Make plot showing differences between the wP and aP groups
```

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).

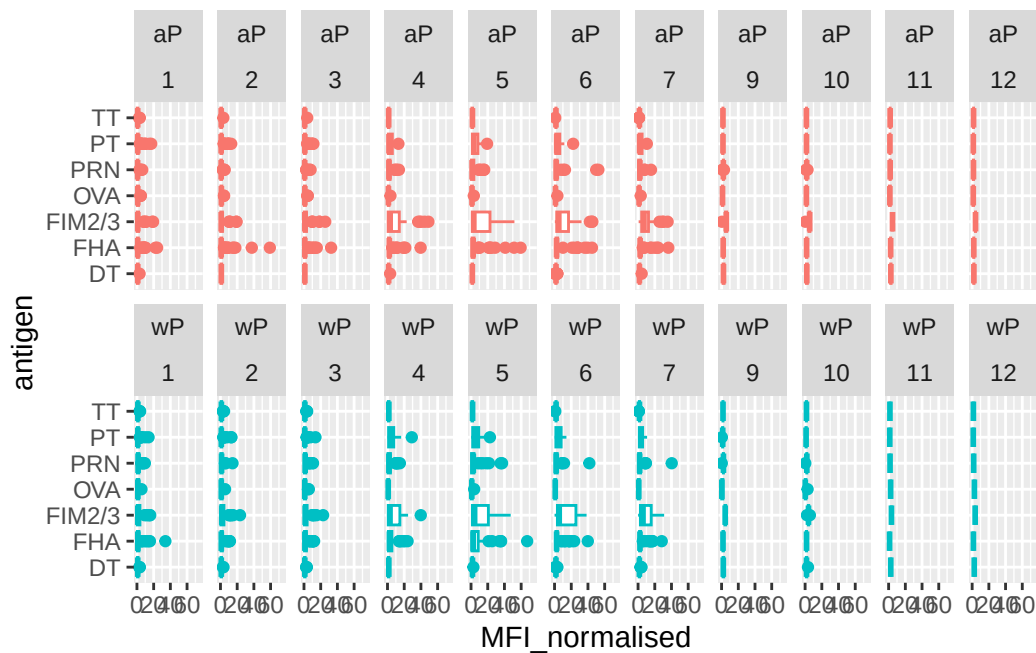


```
# Add infancy-vac to the faceting, for an easier-to-read plot
```

```
igg %>% filter(visit != 8) %>%
```

```
ggplot() +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  xlim(0,75) +
  facet_wrap(vars(infancy_vac, visit), nrow=2)
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).

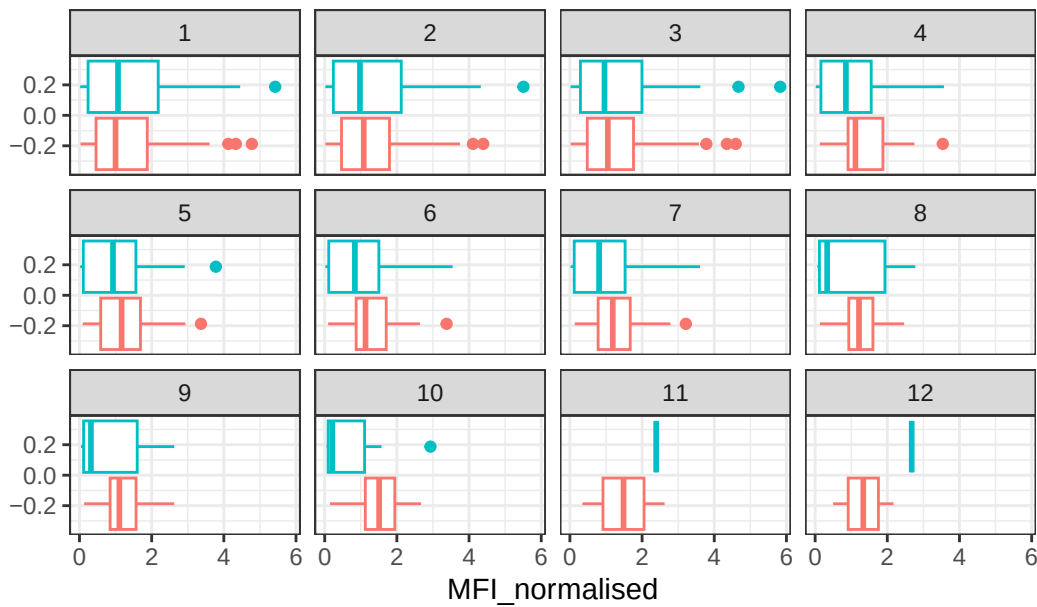


Q15. Filter to pull out only two specific antigens for analysis and create a boxplot for each. You can choose any you like. And the same for FIM2/3.

Boxplot for OVA (negative control):

```
filter(igg, antigen=="OVA") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw() +
  labs(title = "OVA Antigen Levels per Visit (wP group in teal, aP group in red)")
```

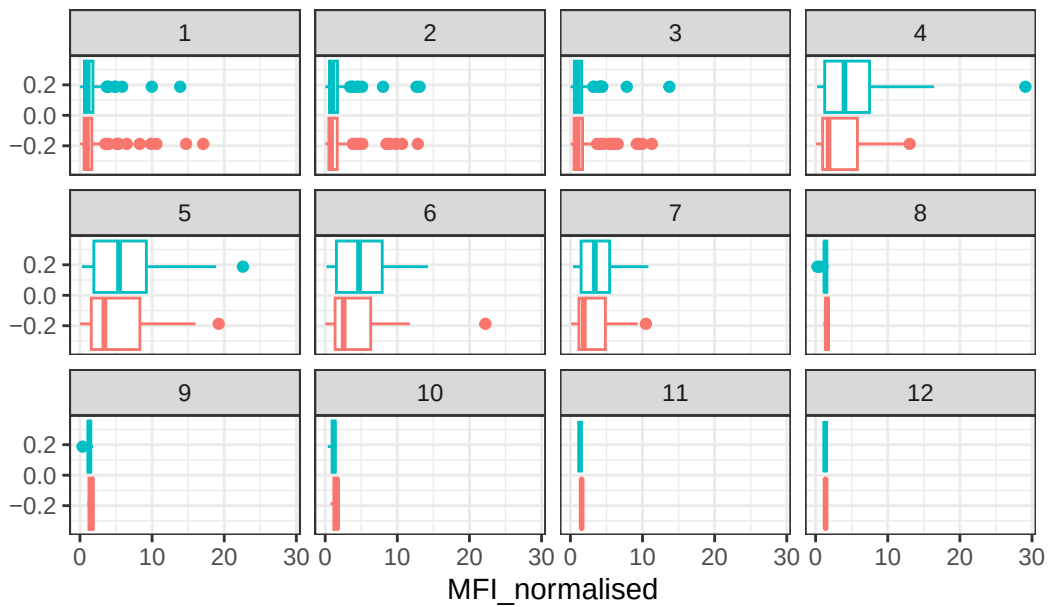
OVA Antigen Levels per Visit (wP group in teal, aP group in red)



Boxplot for PT (pertussis toxin, an aP antigen):

```
filter(igg, antigen=="PT") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw() +
  labs(title = "PT Antigen Levels per Visit (wP group in teal, aP group in red)")
```

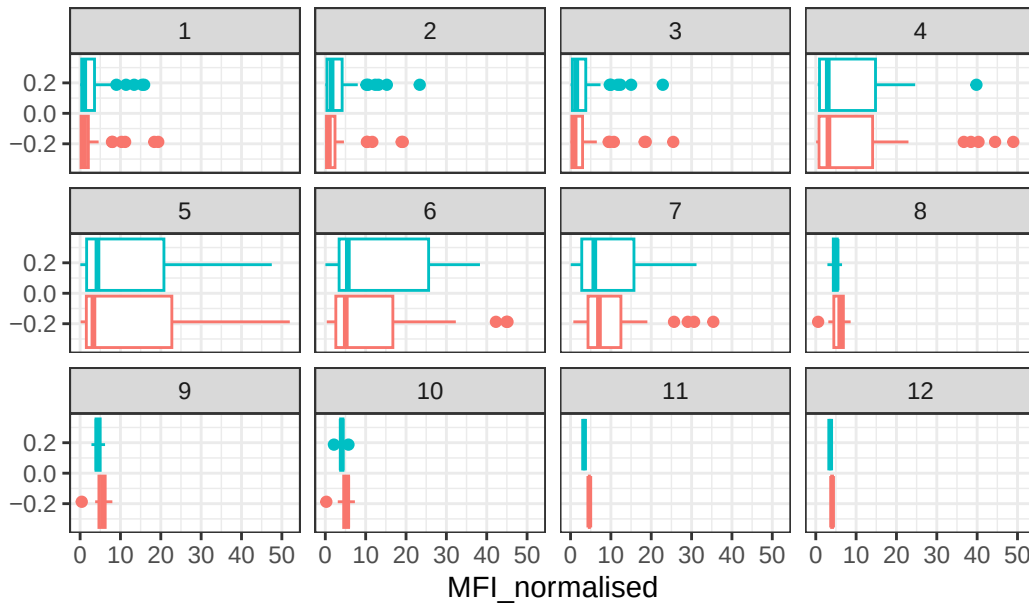

PT Antigen Levels per Visit (wP group in teal, aP group in red)



Boxplot for FIM2/3 (Fimbrial protein 2/3, an aP antigen)

```
filter(igg, antigen=="FIM2/3") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw() +
  labs(title = "FIM2/3 Antigen Levels per Visit (wP group in teal, aP group in red)")
```

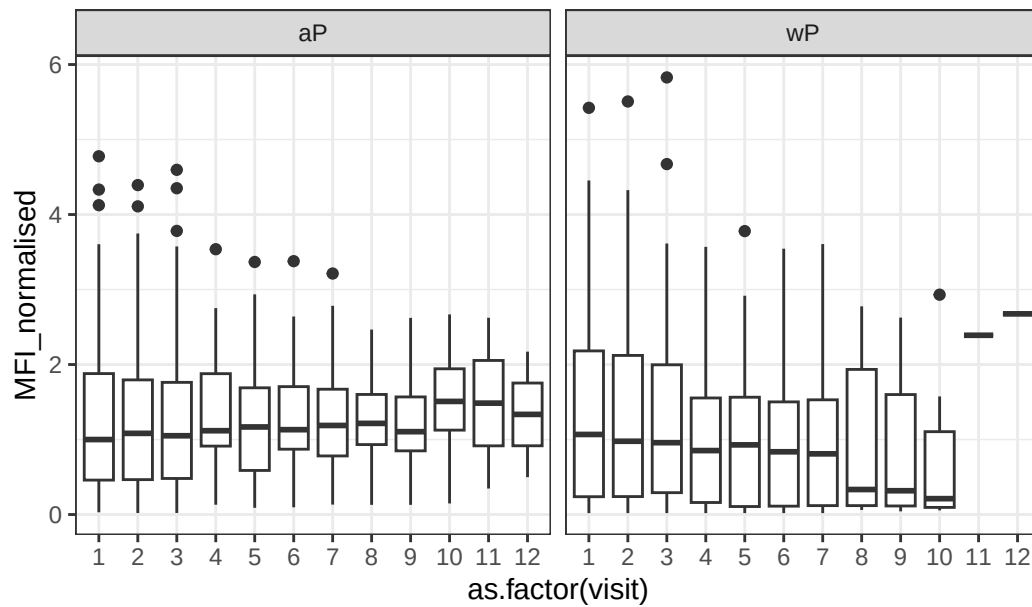
FIM2/3 Antigen Levels per Visit (wP group in teal, aP group in red)



Note: The following, *optional* plots were generated for OVA, PT, and FIM2/3, respectively:

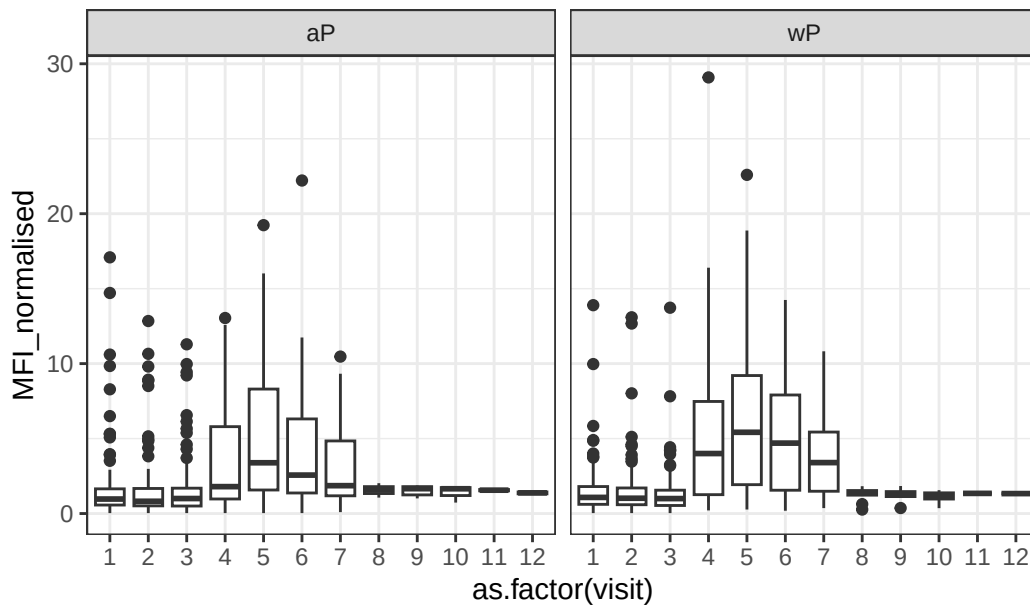
```
filter(igg, antigen=="OVA") %>%
  ggplot() +
    aes(as.factor(visit), MFI_normalised) +
    geom_boxplot(show.legend = FALSE) +
    facet_wrap(vars(infancy_vac)) +
    theme_bw() +
    labs(title = "OVA antigen levels over time, post-Tdap injection")
```

OVA antigen levels over time, post-Tdap injection



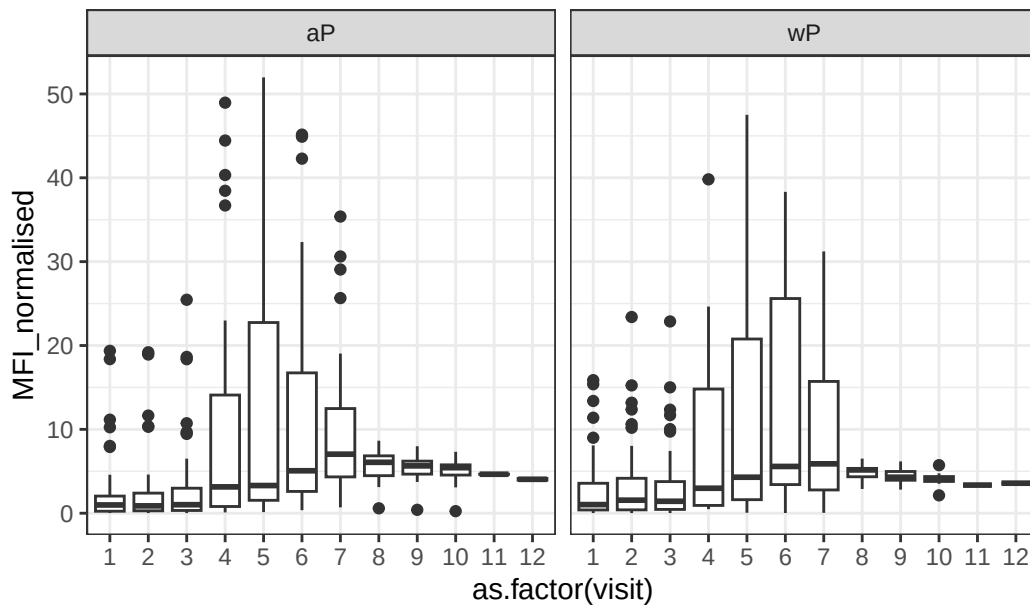
```
filter(igg, antigen=="PT") %>%
  ggplot() +
    aes(as.factor(visit), MFI_normalised) +
    geom_boxplot(show.legend = FALSE) +
    facet_wrap(vars(infancy_vac)) +
    theme_bw() +
    labs(title = "PT antigen levels over time, post-Tdap injection")
```

PT antigen levels over time, post-Tdap injection



```
filter(igg, antigen=="FIM2/3") %>%
  ggplot() +
    aes(as.factor(visit), MFI_normalised) +
    geom_boxplot(show.legend = FALSE) +
    facet_wrap(vars(infancy_vac)) +
    theme_bw() +
    labs(title = "FIM2/3 antigen levels over time, post-Tdap injection")
```

FIM2/3 antigen levels over time, post-Tdap injection



Q16. What do you notice about these two antigens time courses and the PT data in particular?

The OVA (negative control) levels do not increase over time for either the aP or wP groups, staying relatively constant post-Tdap injection. In contrast, the PT (pertussis toxin; an aP antigen) levels are elevated by visit 4, reach a peak at visit 5, and diminish significantly by visit 8. Following a similar trend, the FIM2/3 (another aP antigen) levels are also elevated by visit 4, though they peak at visit 7 and stay elevated (i.e. do not diminish significantly afterwards).

Q17. Do you see any clear difference in aP vs. wP responses?

There does not appear to be any clear difference in the aP versus wP responses, both of which show very similar changes over time (as described in Q16) in PT and FIM2/3 IgG titer levels. Likewise, neither group shows an increase in OVA levels over time, as expected for the negative control.

Note: It is worth mentioning that the PT IgG titer levels appear to reach higher levels in any given visit for the wP group than the aP group. However, this is specific to the PT antigen and is not observed for FIM2/3.

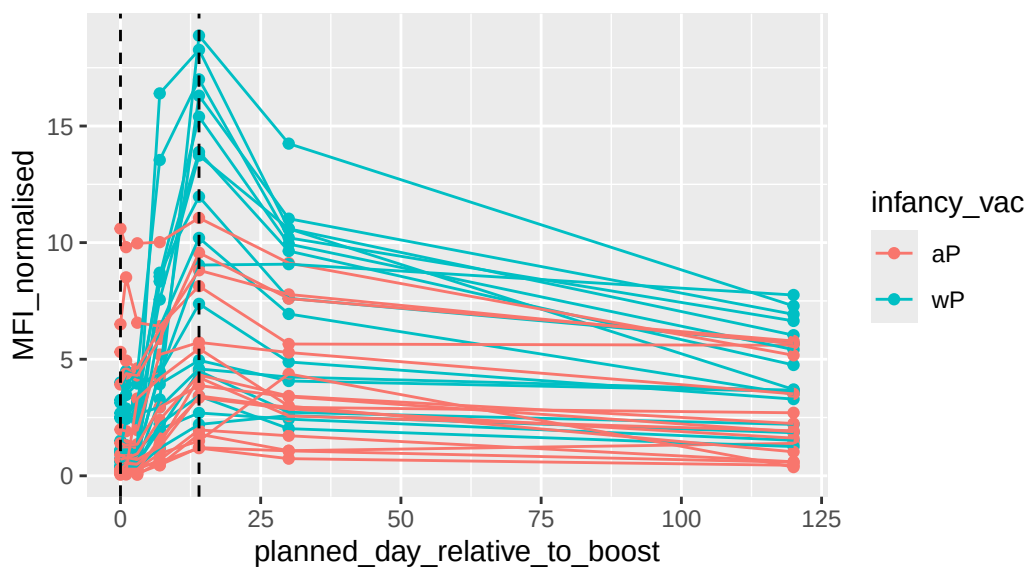
```
# Plot time course data for IgG PT antigen levels, from the 2021 dataset

abdata.21 <- abdata %>% filter(dataset == "2021_dataset")
```

```
abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
    labs(title="2021 dataset IgG PT",
         subtitle = "Dashed lines indicate days 0 (pre-boost) and 14 (apparent peak levels)")
```

2021 dataset IgG PT

Dashed lines indicate days 0 (pre-boost) and 14 (apparent peak levels)



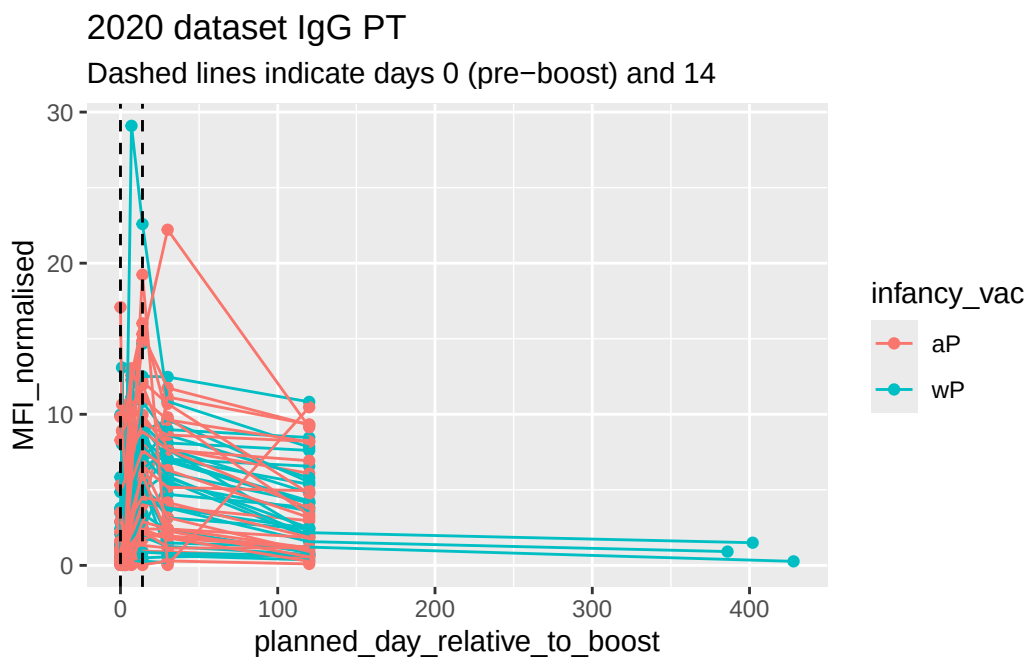
Q18. Does this trend look similar for the 2020 dataset?

```
# Plot time course data for IgG PT antigen levels, from the 2020 dataset

abdata.20 <- abdata %>% filter(dataset == "2020_dataset")

abdata.20 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
```

```
ggplot() +
  aes(x=planned_day_relative_to_boost,
      y=MFI_normalised,
      col=infancy_vac,
      group=subject_id) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept=0, linetype="dashed") +
  geom_vline(xintercept=14, linetype="dashed") +
  labs(title="2020 dataset IgG PT",
       subtitle = "Dashed lines indicate days 0 (pre-boost) and 14")
```



```
# The same plot as above for 2020, taking a closer look at the first 125 days relative to boost

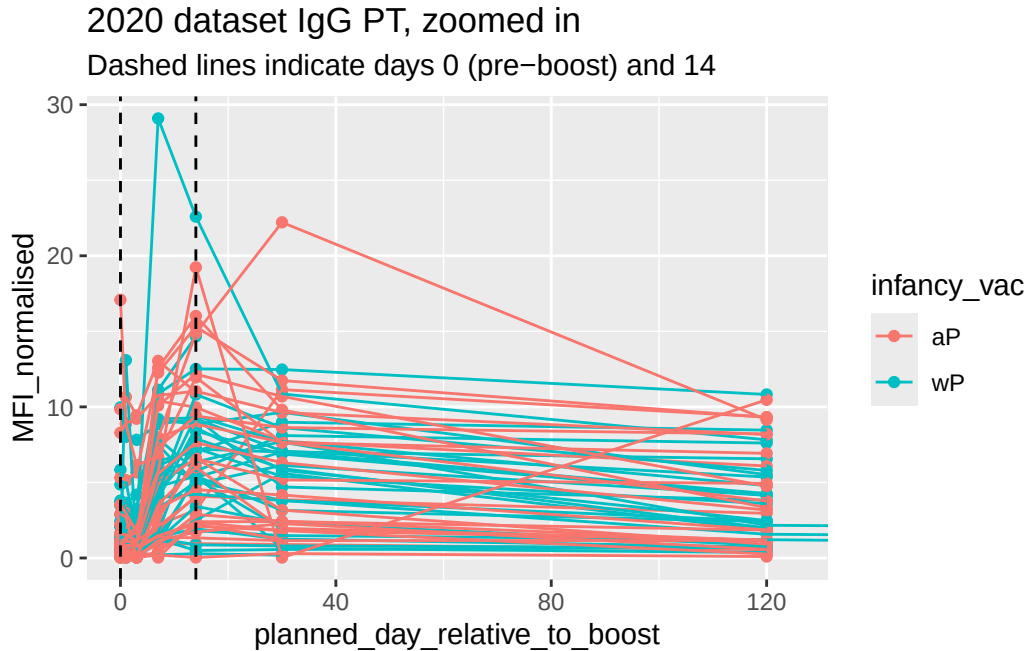
abdata.20 <- abdata %>% filter(dataset == "2020_dataset")

abdata.20 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
```

```

    group=subject_id) +
  geom_point() +
  geom_line() +
  coord_cartesian(xlim = c(0, 125)) +
  geom_vline(xintercept=0, linetype="dashed") +
  geom_vline(xintercept=14, linetype="dashed") +
  labs(title="2020 dataset IgG PT, zoomed in",
       subtitle = "Dashed lines indicate days 0 (pre-boost) and 14")

```



The 2020 dataset does appear to show the same overall trend as the 2021 dataset. In both datasets, IgG PT titer levels seem to generally peak at day 14 relative to receiving the booster injection, and decrease in the days following.

It is worth noting, however, that the 2020 dataset shows more variability than the 2021 dataset (perhaps due to having more data, as seen in Q12), with some peaking—particularly among the wP-vaccinated group—occurring at day 30 post-boost instead. However, the peak IgG PT titer level still, more generally, remains at day 14.

Section 5. Obtaining CMI-PB RNASeq data


```
# Read in available RNA-Seq data for the IGHG1 gene
url <- "https://www.cmi-pb.org/api/v2/rnaseq?versioned_ensembl_gene_id=eq.ENS00000211896.7"

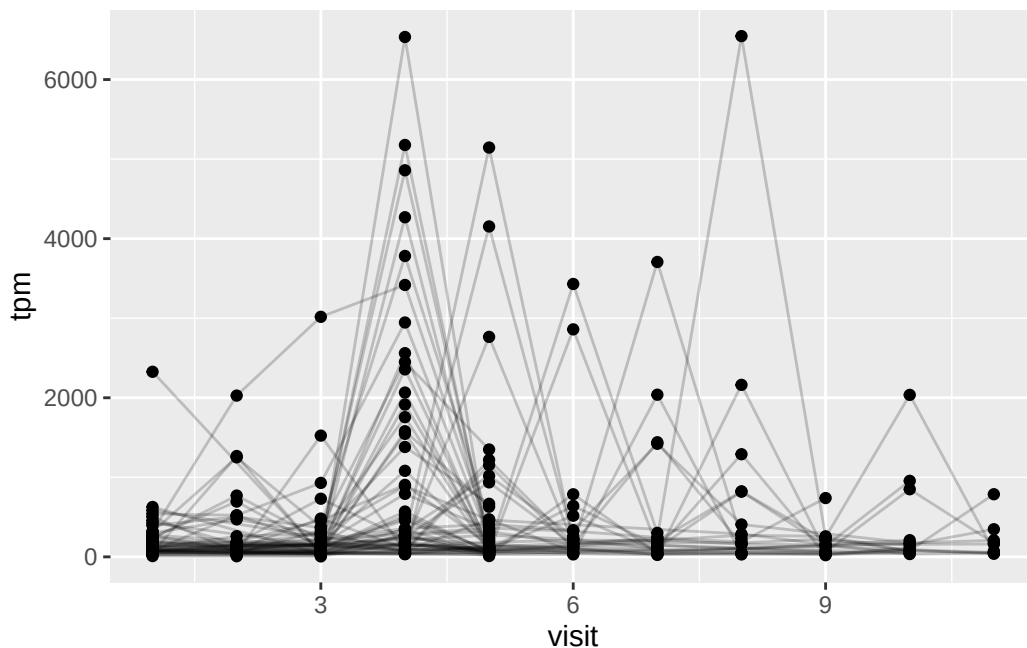
rna <- read_json(url, simplifyVector = TRUE)

# Join the rna expression data with our metadata meta (i.e. sample + specimen data).
ssrna <- inner_join(rna, meta)
```

Joining with `by = join\_by(specimen\_id)`

Q19. Make a plot of the time course of gene expression for IGHG1 gene (i.e. a plot of visit vs. tpm).

```
ggplot(ssrna) +
  aes(visit, tpm, group=subject_id) +
  geom_point() +
  geom_line(alpha=0.2)
```



Q20.: What do you notice about the expression of this gene (i.e. when is it at its maximum level)?

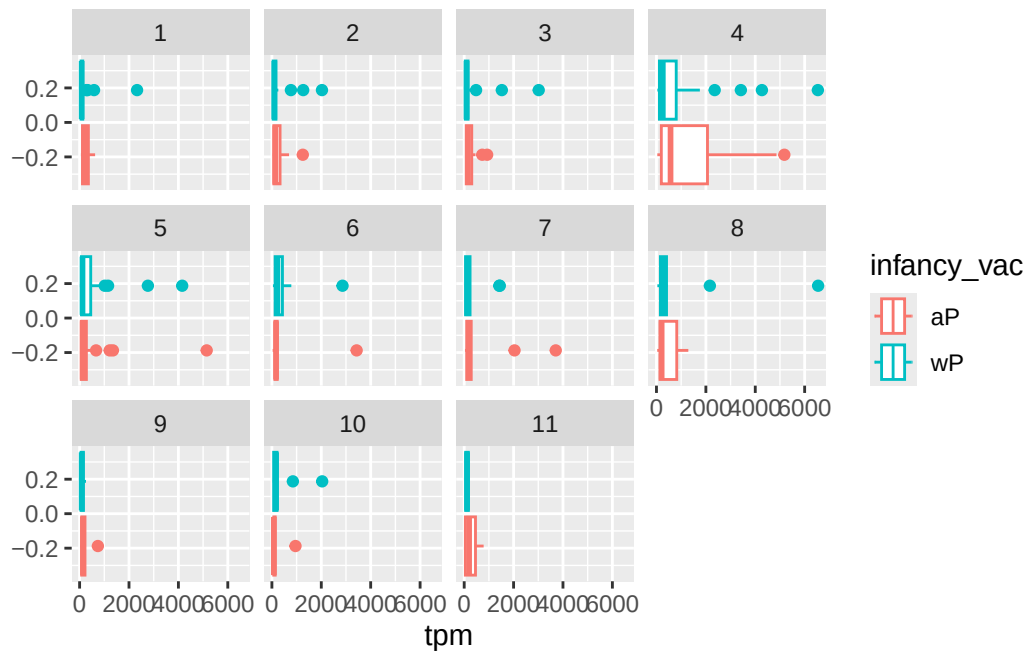
The expression of the IGHG1 gene generally appears to peak at visit 4. There is variability, with some peaking occurring at later visits; however, the overall trend remains that maximum gene expression largely occurs at visit 4.

Q21. Does this pattern in time match the trend of antibody titer data? If not, why not?

Yes, the pattern in time does match. The trend of antibody titer data for IgG PT, taken as an example, peaked around visit 5 (see Q15, 16). It makes sense, then, that peak IGHG1 gene expression generally occurs beforehand, at visit 4: since IGHG1 expression drives production of the IgG1 antibody, it is reasonable and expected for maximum gene expression to precede maximum antibody titer levels.

```
# Color and facet IGHG1 gene expression time course by infancy_vac status
```

```
ggplot(ssrna) +  
  aes(tpm, col=infancy_vac) +  
  geom_boxplot() +  
  facet_wrap(vars(visit))
```



```
# Density plot of IGHG1 gene expression, focusing on visit 4 (no clear wP vs. aP difference)
```

```
ssrna %>%
```

```
filter(visit==4) %>%  
ggplot() +  
  aes(tpm, col=infancy_vac) + geom_density() +  
  geom_rug()
```

